

ECON 0100 | Homework B1 | Demand

I Solutions

Answers and work are shown in the red blocks.

Butterbeer

Preferences for butter beer can be represented by the following demand curve:

$$P_d = 100 - \frac{1}{2}Q_d$$

Prices are in galleons and quantity is in bottles.

Q1 | Quantity Demanded

Use a graph to plot this demand curve, including the quantity demanded at both 40 galleons and 50 galleons.

a) Quantity demanded at 40 galleons: _____

120 bottles.

Plot. Vertical intercept, $Q_d = 0$: $P_d = 100$. Horizontal intercept, $P_d = 0$: $0 = 100 - \frac{1}{2}Q_d \rightarrow \frac{1}{2}Q_d = 100 \rightarrow Q_d = 200$.

$$40 = 100 - \frac{1}{2}Q_d \rightarrow \frac{1}{2}Q_d = 60 \rightarrow Q_d = 120$$

b) Quantity demanded at 50 galleons: _____

100 bottles.

$$50 = 100 - \frac{1}{2}Q_d \rightarrow \frac{1}{2}Q_d = 50 \rightarrow Q_d = 100$$

c) How much did quantity demanded change as the price dropped from 50 to 40? _____

Up by 20 bottles.

■ $\Delta Q_d = 120 - 100 = 20$: quantity demanded rose by 20 bottles as the price fell, the law of demand.

d) What is the marginal benefit at a quantity of 100? _____

50 galleons.

Quantity to price. The height of demand at 100 bottles is the marginal benefit of the hundredth bottle: $P_d = 100 - \frac{1}{2} \cdot 100 = 50$ galleons. It matches b): 50 galleons is the price at which 100 bottles are demanded.

Q2 | Consumer Surplus

Then find and label the consumer surplus at these prices.

a) CS at 40 galleons: _____

3600 galleons.

Consumer surplus is the triangle below demand and above the price, out to the quantity bought from Q1. Its height is the 100-galleon intercept minus the price and its base is the quantity.

$$P = 40 : \quad CS = h \cdot b \cdot \frac{1}{2} = (100 - 40) \cdot 120 \cdot \frac{1}{2} = 60 \cdot 60 = 3600$$

b) CS at 50 galleons: _____

2500 galleons.

$$P = 50 : \quad CS = h \cdot b \cdot \frac{1}{2} = (100 - 50) \cdot 100 \cdot \frac{1}{2} = 50 \cdot 50 = 2500$$

c) How much did consumer surplus change as the price dropped from 50 to 40? _____

Up by 1100 galleons.

$\Delta CS = CS - CS = 3600 - 2500 = 1100$: consumer surplus rose by 1100 galleons, the strip between the two price lines.

Expenditure is the rectangle below the price:
 $40 \cdot 120 = 4800$ at 40 galleons and $50 \cdot 100 = 5000$ at 50.

